



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

UNIVERSITY MANAGEMENT SYPSTEM

A Software Engineering Project Submitted

By

Semester: Spring _24_25		Section: K	Group Number: 03	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	MD. RAIHAN KABIR NIBIR	22-49956-3		
2	SADIA AFRIN	22-49854-3		
3	MOHTASIM-UR-RAHMAN	22-49461-3		
4	ZAHID HOSSAIN	22-49437-3		

The project will be evaluated for the following Course Outcomes

CO3: <i>Select</i> appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation Regarding Process Model Selection	[5Marks]
	Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4: <i>Develop</i> a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	

Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	
CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain	Total Marks	
Taking project responsibility: perform assigned tasks on time independently	[5 Marks]	
Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	
Showing respect and value towards other team member's opinion	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: MD. RAIHAN KABIR NIBIR Student ID: 22-49956-3 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1: Co-developed the System Interface Design (UI/UX), including wireframes, layout, and responsive design elements. ▪ Contribution Description 2: Co-created the Project Schedule, including WBS, Gantt Chart, and conducted COCOMO effort estimation. <u>RAIHAN</u> _____ Signature of the Student
Student Name: SADIA AFRIN Student ID: 22-49854-3 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1: Co-developed the System Interface Design (UI/UX) with an emphasis on usability and accessibility. ▪ Contribution Description 2: Co-created the Project Schedule, including WBS, Gantt Chart, and assisted with COCOMO calculations and interpretations. <u>SADIA</u> _____ Signature of the Student
Student Name: MOHTASIM-UR-RAHMAN Student ID: 22-49461-3 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1: Designed the complete Test Plan, identifying testing scope, objectives, and resources.

- Contribution Description 2: Wrote and executed **detailed test cases** to validate system functionality and documented test results.

MOHTASIM-UR
Signature of the Student

Student Name: **ZAHID HOSSAIN**

Student ID: **22-49437-3**

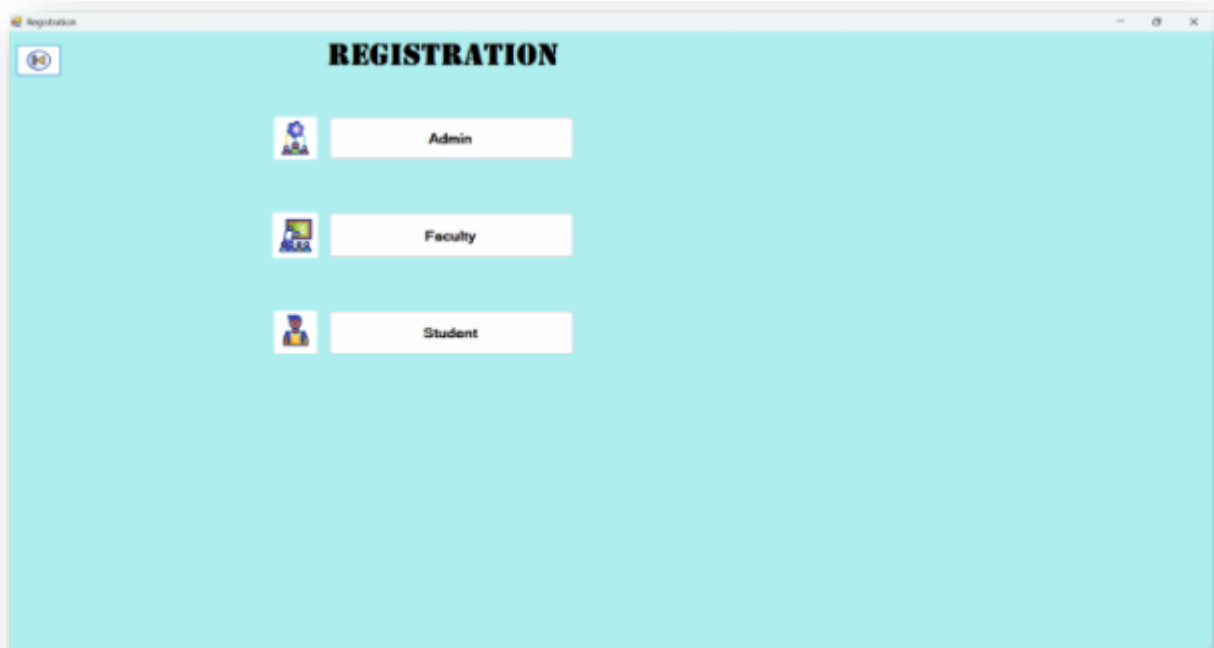
Contribution in Percentage (%): 25%

Contribution in the Project:

- Contribution Description 1: Created the **Risk Management Plan**, identifying potential project risks and mitigation strategies.
- Contribution Description 2: Categorized risks, assigned risk owners, and documented the response and impact levels.

ZAHID
Signature of the Student

System Interface Design (UI/UX):



Form3

ADMIN REGISTRATION

 Enter Code :

☐ Show Code



ENTER

Form4

ADMIN INFORMATION



Name :

User Name :

Password :

Email :

Gender :

Phone Number :

Date Of Birth :

Address :

SUBMIT

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Form8

ADMIN DASHBORAD

ADMIN
FACULTY
STUDENT
DEPARTMENT AND
SCHEDULE
FEES
SALARY
LOG OUT

FACULTY MANAGEMENT

Name : Phone :

Id : Address :

Username :

Password :

Gender : Birth of Date : 2025-02-04

E-mail : Salary :

ADD DELET UPDATE DISPLA

facultyID	Name	Username	Password	DOB	Phone	Email	Gender	Address	Salary
1	Sadia	sadia123	sadia19	2025-02-04	01234567891	sadia@gmail.com	Female	dhaka	200.00
6	Rahan	hufuh	jduhd	2025-02-17	01714333066	uhduh@gmail.com	Male	rang	50000.00

Form8

ADMIN DASHBORAD

ADMIN
FACULTY
STUDENT
DEPARTMENT AND
SCHEDULE
FEES
SALARY
LOG OUT

STUDENT MANAGMENT

Name : Address :

Id :

Username : Birth of Date : Tuesday, February 4, 2025

Password :

Gender : Course :

E-mail : Department :

Phone : Fees :

Teacher :

ADD DELET UPDATE DISPLA

ADMIN DASHBORAD

ADMIN
FACULTY
STUDENT
DEPARTMENT AND
SCHEDULE
FEES
SALARY
LOG OUT

DEPARTMENT AND COURSE MANAGEMENT

Department : Course :

Faculty :

	DepartmentName	CourseName	FacultyID
+	CSE	C++	1
+	CSE	CS	1
+			

	DepartmentName
+	CSE
+	EEE
+	

ADMIN DASHBORAD

ADMIN
FACULTY
STUDENT
DEPARTMENT AND
SCHEDULE
FEES
SALARY
LOG OUT

SCHEDULE MANAGEMENT

EXAM SCHEDULE

Day :

8 AM To 10

10 AM To 12

2 PM To 4 PM

	Day	8 AM TO 10 AM	10 AM TO 12 PM	2 PM TO 4 PM
+	1. Saturday	C++	DUC	ACCOUNTING
+	3. Monday	C++	DUC	ACCOUNTING
+				

CLASS SCHEDULE

Day : Faculty Id :

8 AM To 10

10 AM To 12

2 PM To 4 PM

	Day	8 AM TO 10 AM	10 AM TO 12 PM	2 PM TO 4 PM	FacultyID
+					
+					

Form8

ADMIN DASHBORAD

- ADMIN
- FACULTY
- STUDENT
- DEPARTMENT AND
- SCHEDULE
- FEEES
- SALARY
- LOG OUT

FEES MANAGEMENT

Name : Total Amount :

Id : Due payment :

Course : Paid :

Form8

ADMIN DASHBORAD

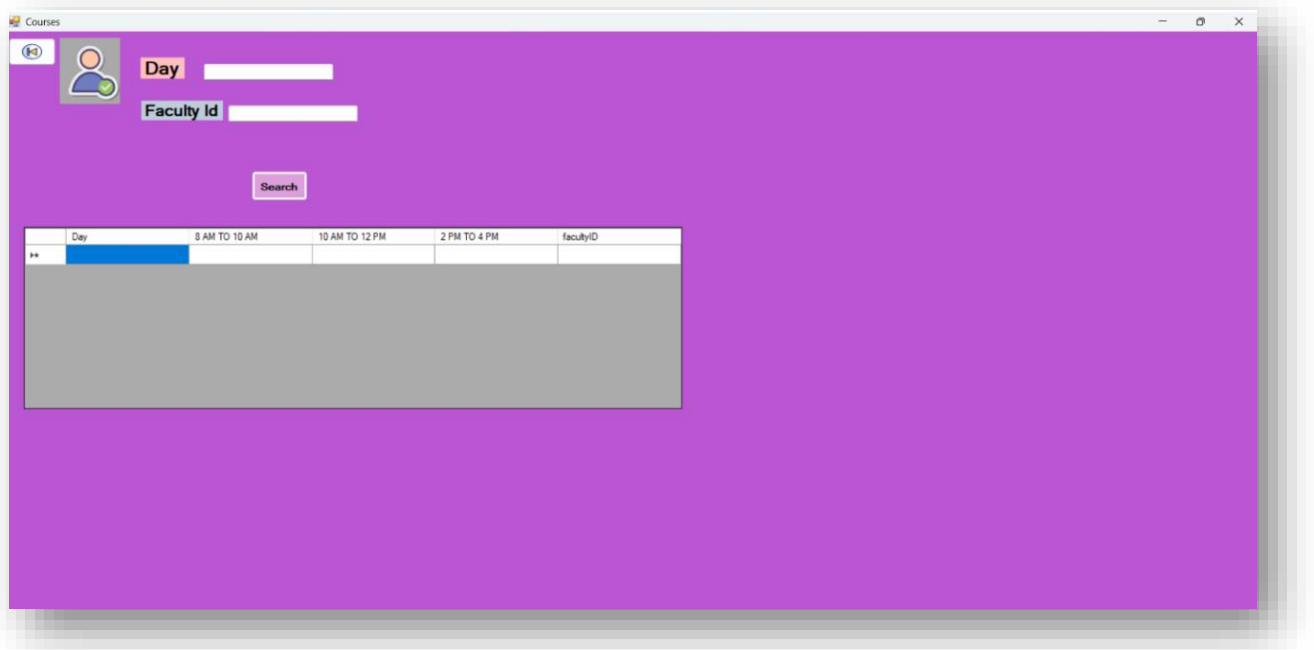
- ADMIN
- FACULTY
- STUDENT
- DEPARTMENT AND
- SCHEDULE
- FEEES
- SALARY
- LOG OUT

SALARY MANAGEMENT

Name : Paid :

Id : Due payment :

Amount :



Assignment

ASSIGNMENT TITLE :

Assignment ID

Course Name:

Submission Tuesday , February 4, 2025 ▾

ADD **UPDATE** **DELETE** **VIEW**

	id	course_name	assignment_title	submission_deadline
▶	1	DLC	EEE	2/12/2025 10:02 PM
*				

StudentReport

STUDENT REPORT

Student **Course Name**

Assignment Marks: **Exam Grades**

Upload Marks **Search Student**

Personal Information



Username

Contact

Address

EMAIL:

Edit **Save**



Dashboard

STUDENT DASHBOARD

STUDENT PROFILE

COURSES

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GRADE PERFORM

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COURSEMATERIAL

FEES STATUS

NOTIFICATIONS

DISCUSSION

LOG OUT

STUDENT PROFILE

Student ID :

Email :

Full Name :

Gender : ☒ Male ☐ Female

User Name :

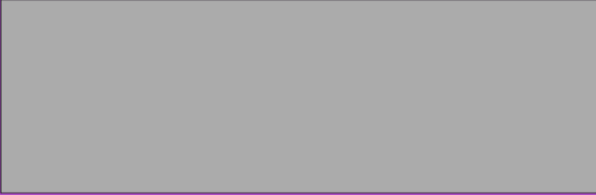
Phone No :

Password :

Address :

Date Of Birth : Tuesday , February 4, 19

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CourseName : Professor :

CourseID : Professor ID :

Department : Credit Hours :

Semester :

ADD **UPDATE** **DELETE**

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EnrollmentID : StudentID :

StudentName : Section :

EnrollDate :

ADD **UPDATE** **SAVE** **DELETE** **DISPLAY** **NEW**

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GRADE PERFORMANCE

Grade ID : CourseID :

Student ID : Grade :

MarksObtained : TotalMarks :

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EXAM SCHEDULE

ExamID : CourseID :

ExamName : ExamType :

ExamDate : Semester :

StartTime : EndTime :

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COURSE MATERIALS

MaterialID : CourseID :

Student ID : UploadDate : Tuesday , February 4, 2025

File : Uploaded By :

Description :

Dashboard

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FEES STATUS

Fee ID : TotalAmount :

Student ID : PaymentDate: Tuesday , February 4, 2025

Paid Amount : DueAmount :

DueAmount :

Dashboard

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NOTIFICATIONS

NotificationID: Student ID :

Title :

DateSent : Tuesday , February 4, 2025 Message :

NotificationType :

Status :

Dashboard

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COMMUNICATION PORTAL

Student ID : Faculty ID :

Name : Name:

Email: Email :

MessageContent Status :

DateStamp: Tuesday , February 4, 2025

TeacherRegistrationForm

Faculty Registration



NAME

Username

password

DOB

Phone

Email

Gender

Addres

StudentRegistration

STUDENT INFORMATION FORM



Name :

Username :

Password :

Email :

Gender :

Phone Number :

Date Of Birth :

Address :

Department :

Test Plan and Test Cases:

Test plan

Objective:

To verify that the developed software system meets all functional and non-functional requirements, ensuring correctness, reliability, usability, and performance.

Scope:

This testing covers all major modules of the system, including login, user management, course registration, result processing, notifications, and administrative controls.

Testing Types:

- Unit Testing – Verifies each function/module in isolation.
- Integration Testing – Tests the flow of data between modules.
- System Testing – Verifies the complete and integrated system.
- Acceptance Testing – Validates the system against user requirements.
- Usability Testing – Ensures the system is user-friendly.
- Security Testing – Checks for vulnerabilities and unauthorized access.

Tools Used:

- Selenium (for automated testing)
- Postman (for API testing)
- JUnit / PyTest / Mocha (depending on development language)
- Manual testing (for UI and end-user behavior)

Test Environment:

- Web Browser: Chrome, Firefox
- OS: Windows 10, Linux (Ubuntu)
- Server: Apache / Nginx
- Database: MySQL / PostgreSQL

Test schedule

Phase	Activity	Start Date	End Date	Responsible Team
Test Planning	Create test plan and test cases	Day 1	Day 3	QA Lead
Environment Setup	Configure test environment & tools	Day 4	Day 5	DevOps / QA
Unit Testing	Developer-level testing of functions	Day 6	Day 10	Developers
Integration Testing	Test interactions between modules	Day 11	Day 13	QA Team
System Testing	Full system test against requirements	Day 14	Day 18	QA Team
Performance Testing	Load and stress testing	Day 19	Day 20	QA + Performance Eng
Security Testing	Test for vulnerabilities and access	Day 21	Day 22	Security Analyst
User Acceptance Testing	Final testing by end-users	Day 23	Day 25	End Users + QA
Bug Fix & Retest	Resolve defects and retest	Day 26	Day 28	Dev + QA
Final Sign-off	Confirm test closure and prepare reports	Day 29	Day 30	Project Manager

Test Resources

Human Resources:

- Test Manager / QA Lead
- QA Testers (Manual & Automation)
- Developers (for unit tests and bug fixing)
- Security Analyst
- UAT Participants (selected end users)
- DevOps Engineer (for environment setup)

Technical Resources:

- Test Environment (Staging Server)
- Browsers: Chrome, Firefox, Edge
- Automation Tools: Selenium, JUnit, PyTest
- API Testing: Postman
- Version Control: Git
- Bug Tracking Tool: Jira / Trello / Excel Sheet

Test Deliverables:

1. Test Plan Document
2. Detailed Test Case Document
3. Test Scripts (if automated)
4. Test Execution Reports
5. Bug/Defect Reports
6. Test Summary Report
7. Sign-off Document from Stakeholders
8. UAT Feedback & Resolution Log
9. Performance and Security Test Report

Entry Criteria:

- All modules are developed and integrated.
- Test environment is set up.
- Test cases are reviewed and approved.

Exit Criteria:

- All major test cases are executed.
- No critical bugs are unresolved.
- UAT is completed with sign-off.

Roles and Responsibilities

Role	Responsibilities
Test Manager	Oversees the entire testing process, approves plan and ensures delivery timelines.
QA Lead	Prepares test plan, assigns tasks, ensures quality and consistency.
QA Testers	Create and execute test cases, log defects, verify bug fixes.
Automation Engineer	Develops automated scripts for regression and load testing.
Developers	Perform unit tests, fix bugs reported by QA, assist in integration testing.
Security Analyst	Conducts security testing and audit reports.
DevOps Engineer	Sets up and maintains the test environment and deployment pipeline.
UAT Participants	Represent users and validate system functionality meets expectations.
Project Manager	Coordinates between teams, signs off on final test reports and deliverables.

Test case

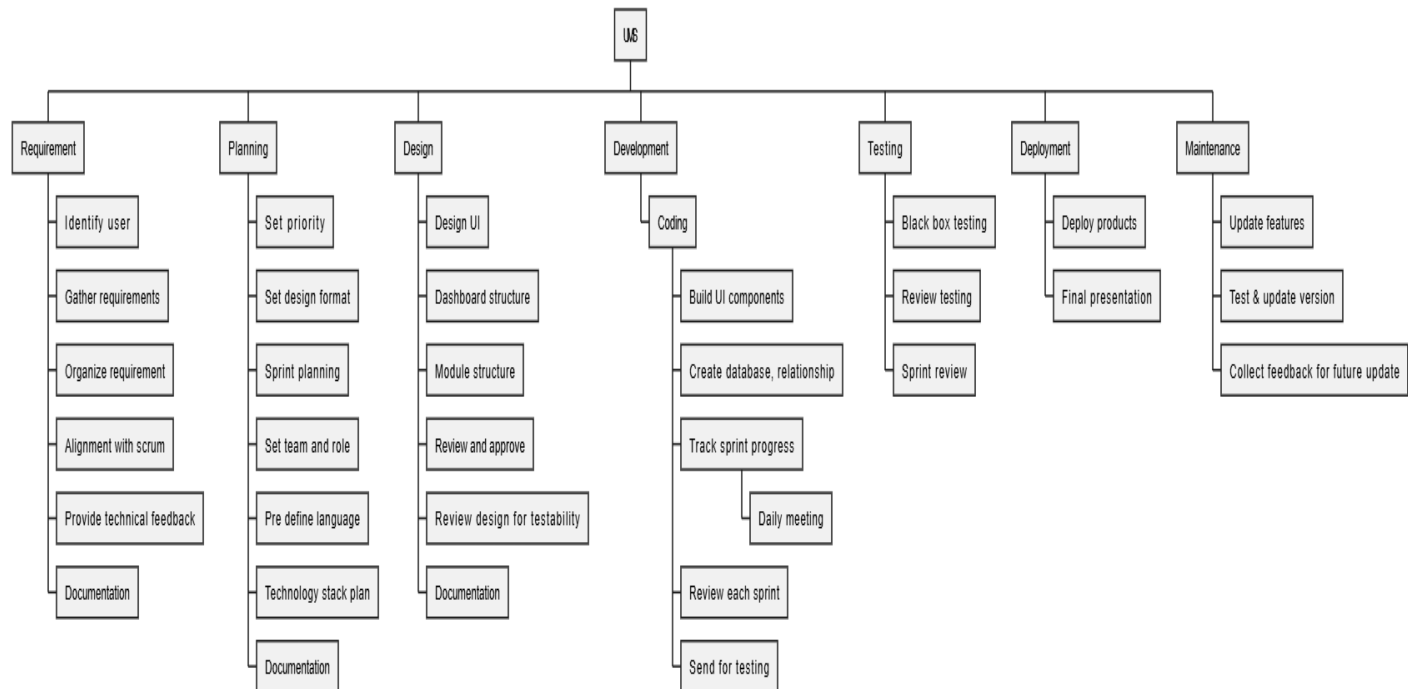
Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC-001	User login with valid credentials	Valid username/password	Dashboard displayed	As expected	Pass
TC-002	User login with invalid credentials	Invalid password	Error message	As expected	Pass
TC-003	Register for a course	Select course and submit	Registration confirmation	As expected	Pass

TC-004	Register same course twice	Repeat registration	Error/Already registered message	As expected	Pass
TC-005	View results	Logged-in student clicks 'View Results'	Show result list	As expected	Pass
TC-006	Admin adds new course	Input valid course details	Course added to list	As expected	Pass
TC-007	Unauthorized access to admin panel	User without admin rights	Access denied	As expected	Pass
TC-008	Notification delivery	Publish notice	Notice received via email/SMS	As expected	Pass
TC-009	Session timeout	Leave idle for 30 min	Auto logout	As expected	Pass
TC-010	System load during exam time	100+ concurrent logins	System handles load without crash	As expected	Pass
TC-011	Password reset	Valid email provided	Reset link sent	As expected	Pass
TC-012	Data validation	Enter invalid email/phone	Error message shown	As expected	Pass
TC-013	Internet failure during exam	Disconnect during submission	Auto-save + retry	As expected	Pass
TC-014	Database failure simulation	Shutdown DB	Error handling / Retry	Handled gracefully	Pass
TC-015	Export results	Admin clicks export	CSV/Excel downloaded	As expected	Pass
TC-016	UI responsiveness	Open app on mobile & tablet	Layout adjusts properly	As expected	Pass
TC-017	Cross-browser compatibility	Open app on Chrome, Firefox, Edge	UI looks consistent	As expected	Pass

TC-018	SQL Injection attempt	'OR 1=1;--' in login field	Access denied / input rejected	As expected	Pass
TC-019	XSS vulnerability	<script>alert('x')</script>	Script blocked or sanitized	As expected	Pass
TC-020	Broken link detection	Click all footer/menu links	All pages open correctly	As expected	Pass
TC-021	Load testing	1000 users accessing at once	No crash or timeout	As expected	Pass
TC-022	Backup and recovery	Simulate crash and restore	System restores correctly	As expected	Pass
TC-023	Password strength enforcement	Try weak password	Error message shown	As expected	Pass
TC-025	Email verification	Register new user	Verification email sent	As expected	Pass

Project Schedule & Effort Estimation:

WBS



COCOMO:

- **Project Type:** Semi-detached (intermediate complexity, size, and experience).
- **Estimated Size:** 17,000 Lines of Code (LOC).

1. Estimated Effort (PW - Person-weeks):

- **Formula Used:** $\text{Effort} = a * (\text{KLOC})^b$
 - Where:
 - $a = 3$ (COCOMO constant for semi-detached projects)
 - $\text{KLOC} = 17,000 / 1,000 = 17$ (Thousands of Lines of Code)
 - $b = 1.12$ (COCOMO exponent for semi-detached projects)
- **Calculation:** $\text{PM} = 3 \times (17)^{1.12}$

$$= 3 \times 23.883 \text{ PM}$$

$$= 71.651$$

- **Rounded Result:** Approximately **72 Person per week**

2. Estimated Development Time (DW - Development Weeks):

- **Formula Used:** Development Time = $c * (\text{Effort})^d$
 - Where:
 - $c = 2.50$ (COCOMO constant for semi-detached projects)
 - Effort = 71.651 (Calculated Person-weeks)
 - $d = 0.35$ (COCOMO exponent for semi-detached projects)
- **Calculation:** $DM = 2.50 \times (71.651)^{0.35}$
 $= 2.50 \times 4.459$
 $= 11.148$
- **Rounded Result:** Approximately **11.15 Weeks** (or roughly 12 weeks when rounded up to a whole number)

3. Required Number of People (ST - Staffing/Team Size):

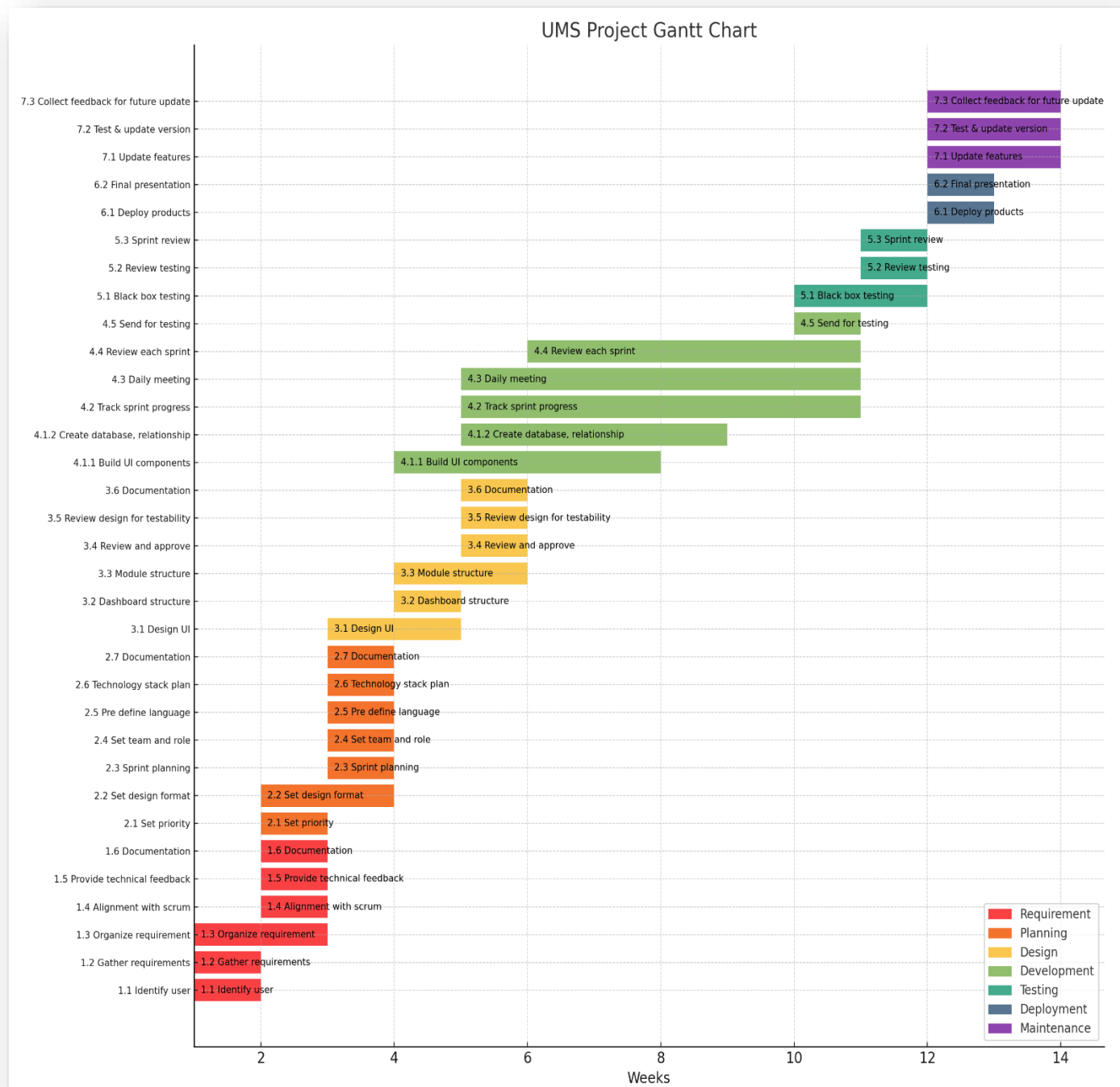
- **Formula Used:** Required Number of People = Effort / Development Time
 - Where:
 - Effort = 71.651 Person-Weeks
 - Development Time = 11.150 Weeks
- **Calculation:** $ST = 71.651 / 11.150$
 $= 6.426$
- **Rounded Result:** Approximately **7 people**

Overall Summary:

Based on the calculations provided for a 17,000 LOC semi-detached software project:

- **Total Estimated Effort:** ~72 Person-Weeks
- **Estimated Development Time:** ~11.15 Weeks
- **Average Team Size:** ~7 people

Gantt Chart:



Risk Management Plan:

ID	Risk	Impact	Response	Risk Level	Owner	Notes	Type of Risk
R-01	Server may stop during online exams	3(High)	Keep backup servers, check system regularly	High (75%)	IT Admin	Do maintenance when no exams	Technology
R-02	Student data may be stolen (hacked)	3(High)	Use strong password and lock system	High (80%)	Security Officer	Scan system for problems often	Security
R-03	System may crash during course registration	3(High)	Test system before busy time	Medium (55%)	IT Team Leader	Try fake load before real students use	Technology
R-04	Course info may be entered wrong	2(Medium)	Double-check and confirm with department head	Medium (50%)	Academic Admin	Add a check before final save	User Data
R-05	Staff may not know how to use the system	2(Medium)	Give training every time system changes	Low (25%)	HR / IT Trainer	Ask staff what help they need	Staff Training
R-06	Internet may stop during online exam	3 (High)	Use backup internet and auto-save answers	High (70%)	Network Admin	Use offline mode if needed	Technology
R-07	Someone may enter the system without right	3 (High)	Give access only to correct people	High (75%)	System Admin	Check who logs in every week	Security
R-08	Students may miss important notices	2(Medium)	Send messages through SMS, app, and email	Medium (45%)	Communication Officer	Use more than one way to tell them	Communication
R-09	Results may come out late	2(Medium)	Fix a clear calendar with deadlines	Medium (50%)	Registrar Office	Add buffer days for safety	Admin Work
R-10	Update may break system features	2(Medium)	Test updates before using them	Low (30%)	Developer Team	Keep backup of old version	Technology

ID	Risk	Impact	Response	Risk Level	Owner	Notes	Type of Risk
R-11	Wrong attendance data may be saved	2(Medium)	Use ID cards or fingerprint to mark attendance	Medium (45%)	Attendance Officer	Compare system and manual record	Admin Work
R-12	ID cards may come late from vendor	2(Medium)	Talk to vendor early and track delivery	Low (20%)	Purchase Officer	Add delivery tracker in system	Vendor Problem

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					

Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	Some errors in spelling and grammar. Some problems of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	Few errors in spelling and grammar. Presents most of the details in a logical flow of organization in definition, details, and example.	Project report is complete and No errors in spelling and grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marks Distribution (Maximum 3X5=15)					
Marking Criteria	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	

Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

CO5 [PO-i-2]: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in computer science and engineering domain.

Assessment Attribute/Criteria	Missing/ Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Taking responsibility	Does not perform assigned tasks; often misses meetings and, when present, does not have anything constructive to say; relies on others to do the work;	Partially performs all assigned tasks; attends meetings irregularly and occasionally participates and hence not reliable;	Performs all assigned tasks; attends meetings regularly and usually participates effectively. generally reliable;	Performs all tasks very effectively; attends all meetings and participates enthusiastically; very reliable.
Contributions	Never provides useful ideas when participating in a group discussion	Rarely provides useful ideas when participating in a group discussion	Sometimes provides useful ideas when participating in a group discussion	Routinely provides useful ideas when participating in a group discussion
Collaboration and Ability to Compromise	Not cooperative, unable to compromise and disrupts the team process.	Sometimes cooperative, and rarely displays a positive attitude.	Usually cooperative, able to compromise and generally display positive attitude.	Always cooperative. Willingness to compromise. Always display positive attitude.

Valuing other team members (Working with others)	Often argues with teammates; doesn't let anyone else talk; occasional personal attacks and "put-downs"; wants to have things done his way and does not listen to alternate approaches.	Seldom listens to others' points of view; occasionally behaves in an oppressive manner; tries to force their own ideologies on other.	Generally, listens to others' points of view; always uses appropriate and respectful language; tries to make a definite effort to understand others' ideas.	Always listens to others and their ideas; helps them develop their ideas while giving them full credit; always helps the team reach a fair decision.
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